

Investigating Resonances in Light Nuclei with ^3He -Induced Reactions

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Motivation

- Particle-unbound resonances in light nuclei provide insight into nuclear structure and provide sensitive benchmarks for nuclear structure theories
- An integrated resonance-decay and charged-particle spectroscopy methodology enables reconstruction of resonance energies and decay widths by using invariant-mass analysis
 - For data where only one resonance-decay particle is detected, the missing particle can be reconstructed using ejectile information

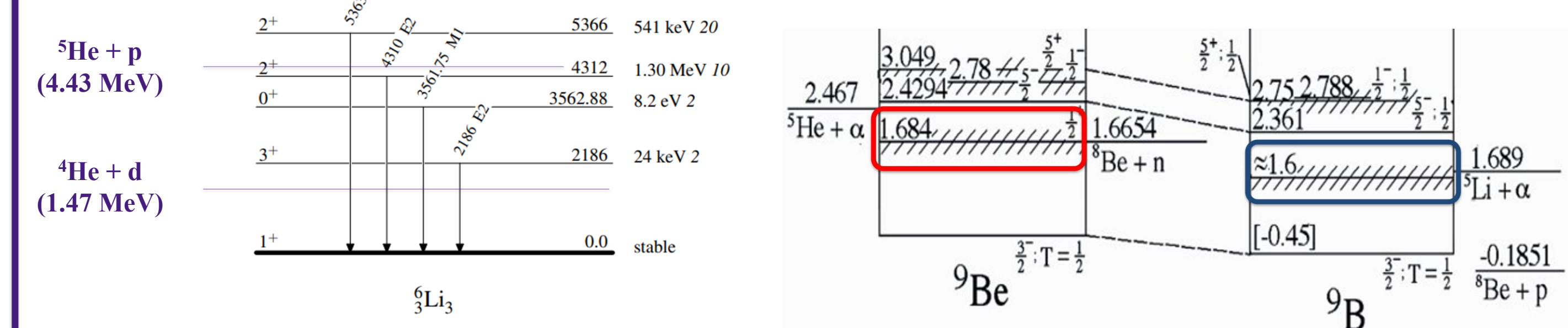


Figure 1: (Left) Level scheme for ^6Li , adapted from [1]. The 3^+ state lies above the alpha separation energy and thus is unstable to alpha decay. The 5.36 MeV 2^+ , $T=1$ state is the IAS to the ground state of ^6Be and ^6He . (Right) Level scheme for mirror nuclei ^9Be and ^9B . The well-known $1/2^+$ state in ^9Be [2] is shown in red and the mirror state in ^9B is shown in blue.

Experimental Details

- An experiment was conducted at the John D. Fox Superconducting Linear Accelerator Laboratory at Florida State University
 - LiF target (75- $\mu\text{g}/\text{cm}^2$, 20- $\mu\text{g}/\text{cm}^2$ C backing), $E_{\text{lab}} = 7.5$ MeV
 - Reaction ejectile measured by SE-SPS in coincidence with resonance-decay particles at backwards angles in the Silicon Array for Branching Ratio Experiments (SABRE) [3]
 - $^7\text{Li}(^3\text{He}, \alpha)^6\text{Li}$ reaction included in dataset with 2- and 1-particle SABRE events
- In the future, an experiment will be conducted at the Cyclotron Institute at Texas A&M University
 - ^3He gas target with stable ^7Li and radioactive ^7Be beams
 - $^7\text{Li}(^3\text{He}, p)^9\text{Be}$ and $^7\text{Be}(^3\text{He}, p)^9\text{B}$ reactions in inverse kinematics
 - Analysis of the same resonance with the same reconstruction methodology with different reaction kinematics

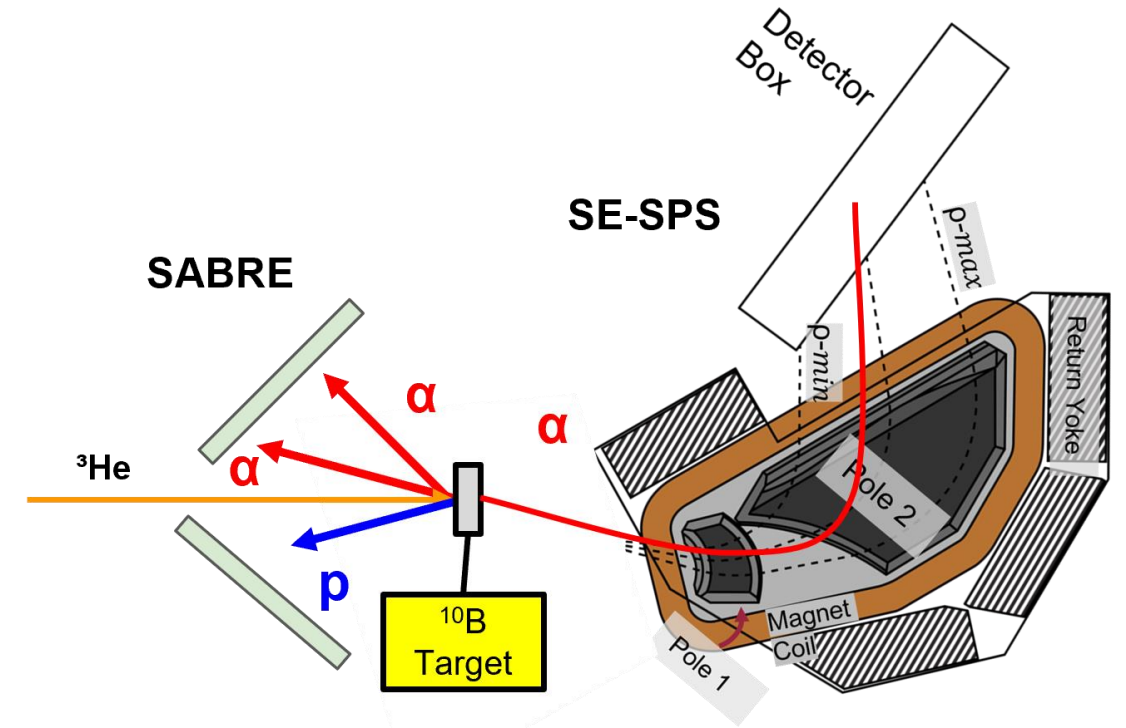


Figure 2:



Figure 4: Photograph of the LSU Gas Target 2.0 (commissioned 2026). QR code links to poster with more details.

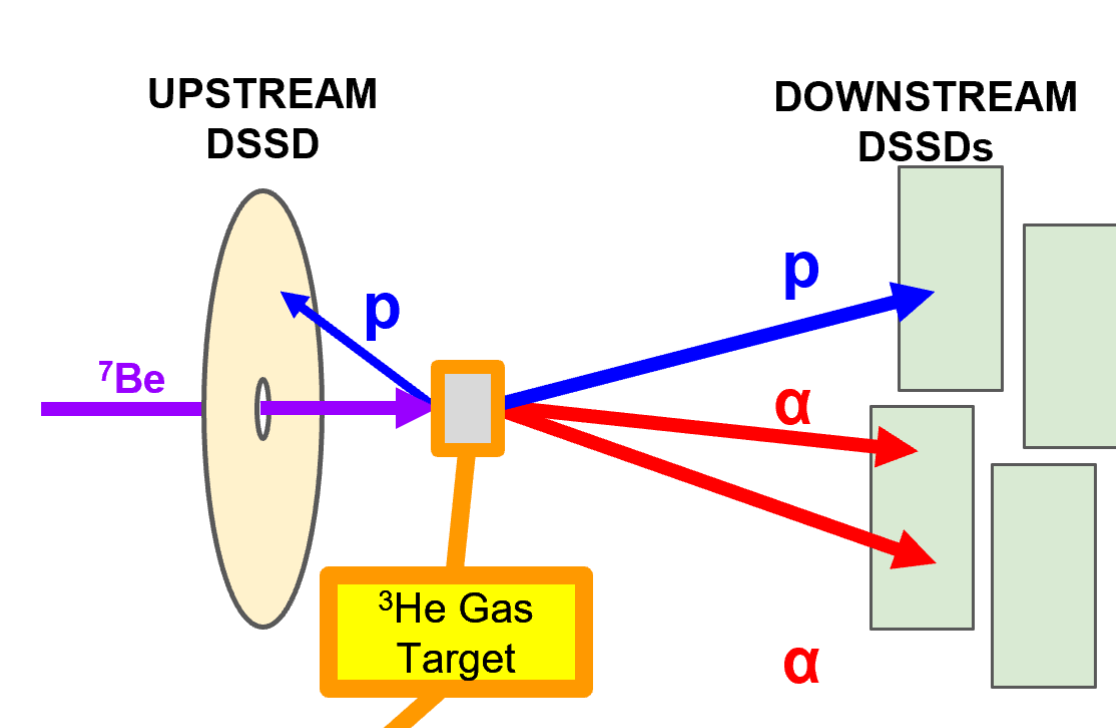


Figure 3:

Invariant Mass Reconstruction

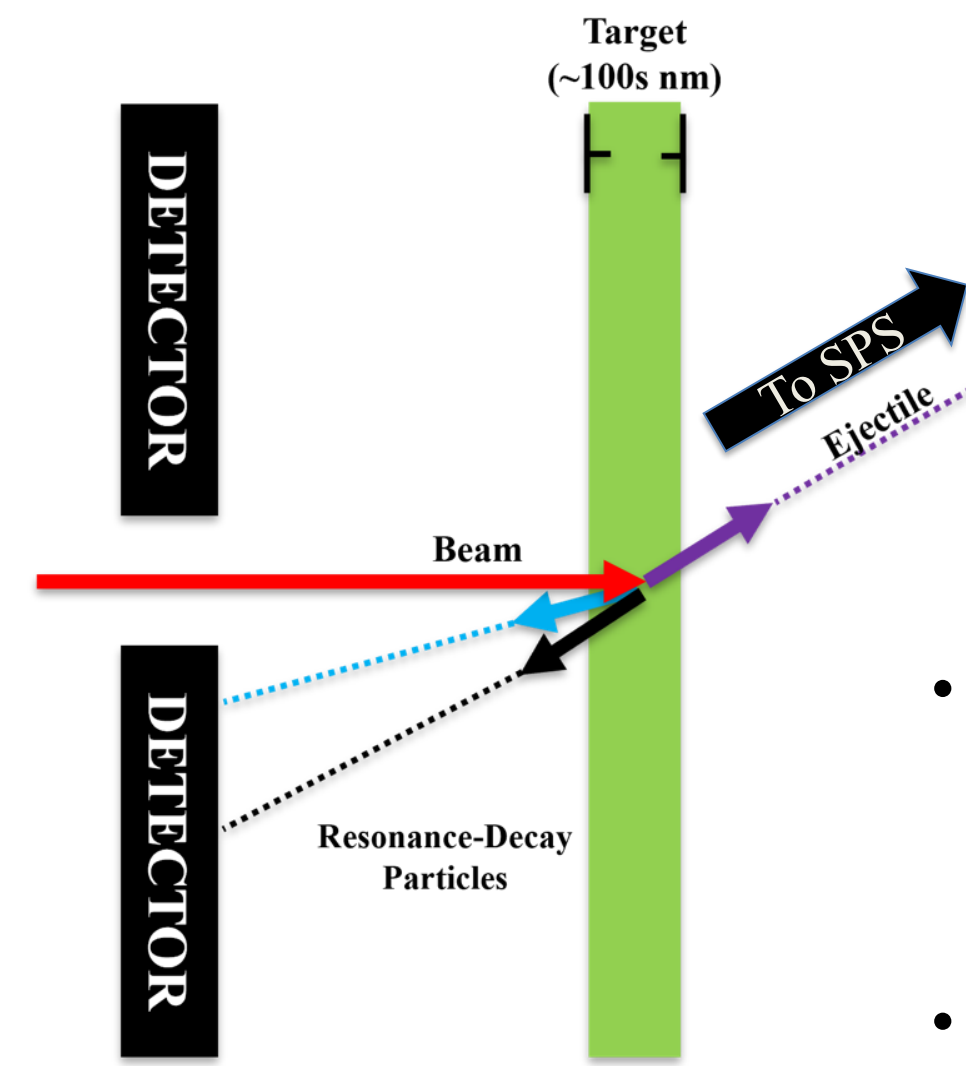


Figure 5: A cartoon of a nuclear reaction populating a particle-resonance in the recoil nucleus. A beam (red arrow) impinges upon a target (green) and a nuclear reaction occurs with the ejectile emitted at forward laboratory angles. The particle-resonance decays via resonance-decay particles (blue, black arrows) and may

Detect Both Resonance Decay Particles (N Case)

$$P_{\text{Recoil}} = \sum_i^N P_i$$

$$P_{\text{Missing}} = P_{\text{Beam}} + P_{\text{Target}} - P_{\text{Ejectile}} - \sum_i^{N-1} P_{\text{Observed},i}$$

Detect Both Resonance Decay Particles ($N-1$ Case)

$$P_{\text{Recoil}} = P_{\text{Missing}} + \sum_i^{N-1} P_{\text{Observed},i}$$

- Invariant mass reconstruction can be applied to nuclear reactions with N -body final states where all resonance-decay particles (N) or all-but-one resonance-decay particles ($N-1$) and the reaction ejectile are detected
- High-resolution charged-particle spectroscopy on reaction ejectile provides reaction kinematics information for $N-1$ cases

Extract excitation energy using reconstructed invariant mass (M)

$$E_{\text{Ex}} = M(P_{\text{rec}}) - M_{\text{gs}} \text{ where } M = \sqrt{E^2 - p^2}$$

$^7\text{Li}(^3\text{He}, \alpha)^6\text{Li}$

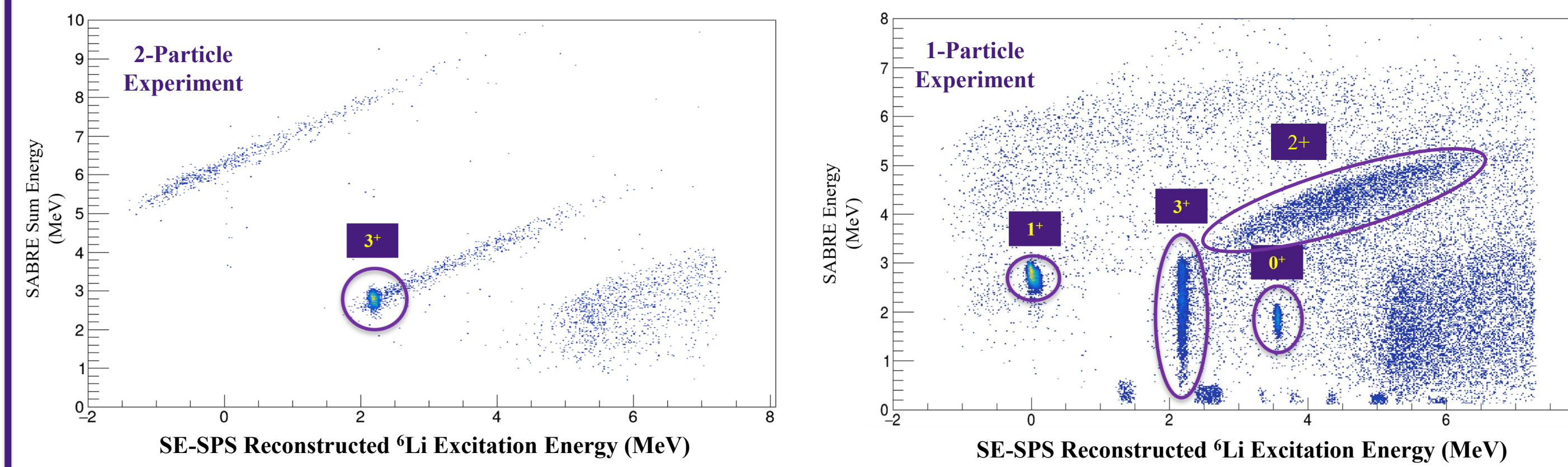


Figure 6: The SABRE sum energy versus the SPS reconstructed ^6Li excitation energy for events with 2 particles in SABRE (left) and for events with 1 particle in SABRE (right).

$^7\text{Li}(^3\text{He}, \alpha)^6\text{Li}_{2.186} \rightarrow \alpha + d$ ($N=2$)

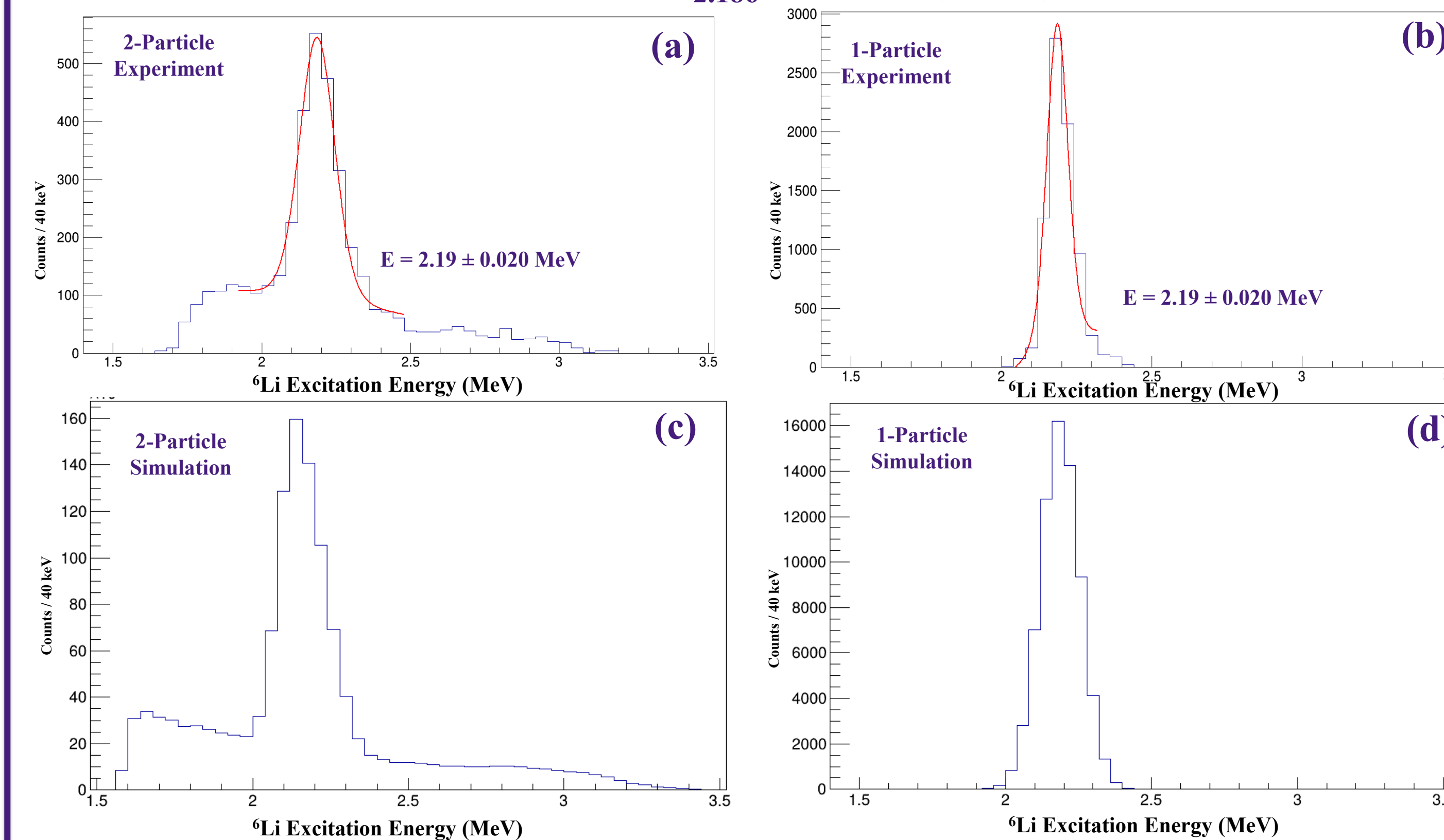


Figure 7: Invariant mass reconstruction of 3^+ state from SABRE 2-particle data (a) and SABRE 1-particle data (b). For comparison, spectra from simulated events where both (c) resonance-decay particles are detected and where only one (d) is detected.

- Particle identification is not necessary to reconstruct the resonance
- All mass-assignment permutations of a particular decay assumption must be tried with the correct assignments resulting in a peak and the incorrect assignments resulting in a broad background

$^{10}\text{B}(^3\text{He}, \alpha)^9\text{B}$

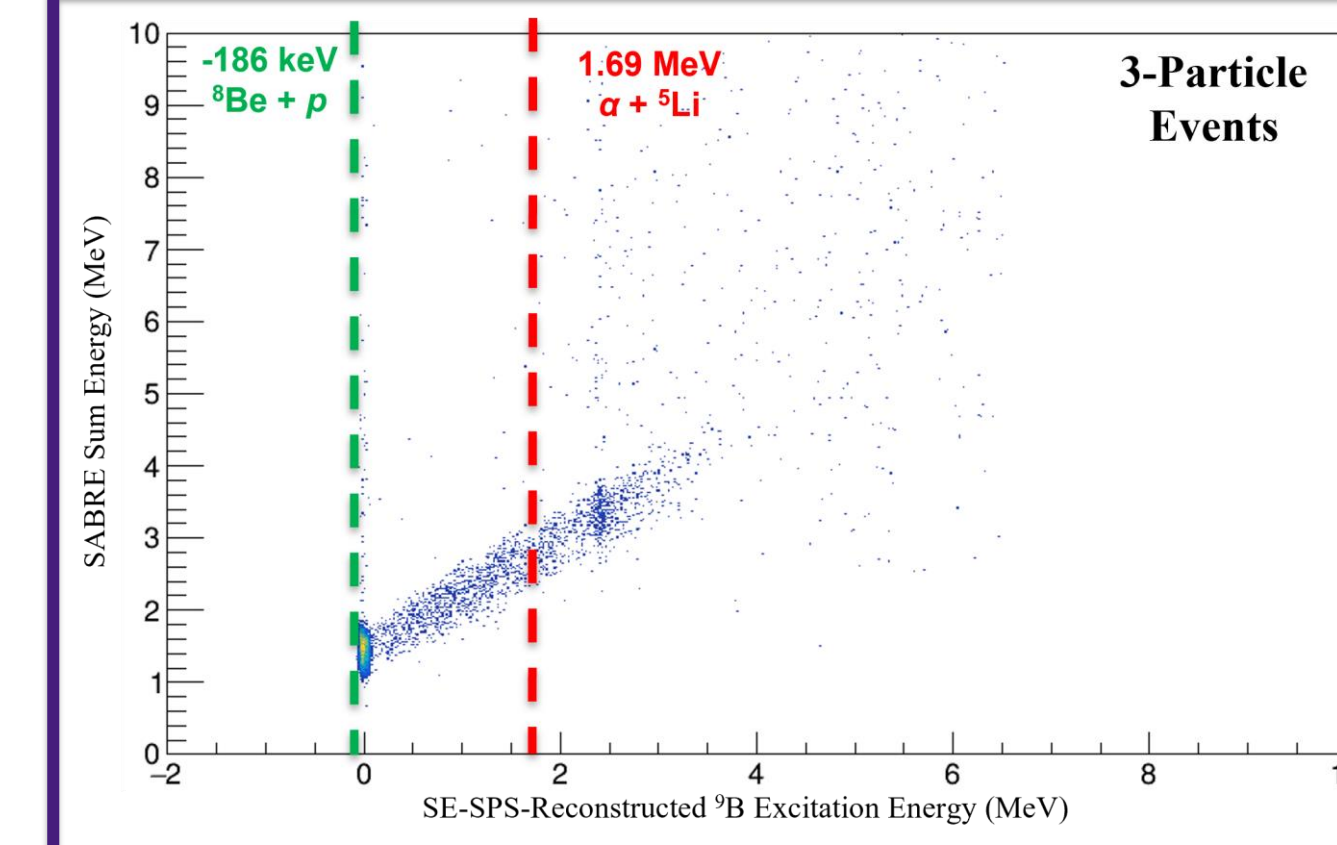


Figure 8 (above): The SABRE sum energy versus the SPS-reconstructed ^9B excitation energy for events with 3 particles in SABRE

- ^9B decays to a 3-particle final state ($p + \alpha + \alpha$) regardless of the intermediate decay ($N=3$)
- Invariant mass of correct decay assumptions result in a peak at the resonance energy
- Invariant mass of incorrect decay assumption spreads out to nonphysical continua

Figure 9 (Below): Excitation energy of the assumed intermediate decay plotted against the SPS-reconstructed excitation energy of the ^9B recoil. The $^{10}\text{B}(^3\text{He}, \alpha)^9\text{B}$ reaction was simulated (10^6 total events) populating both the $E=2.345$ MeV, $J\pi=5/2^-$ and $E=2.751$ MeV, $J\pi=5/2^+$ states in ^9B and decaying through $^9\text{B}^* \rightarrow p + ^8\text{Be}_{\text{gs}} \rightarrow \alpha + \alpha$ (left) and $^9\text{B}^* \rightarrow \alpha + ^5\text{Li}_{\text{gs}} \rightarrow p + \alpha$ (right).

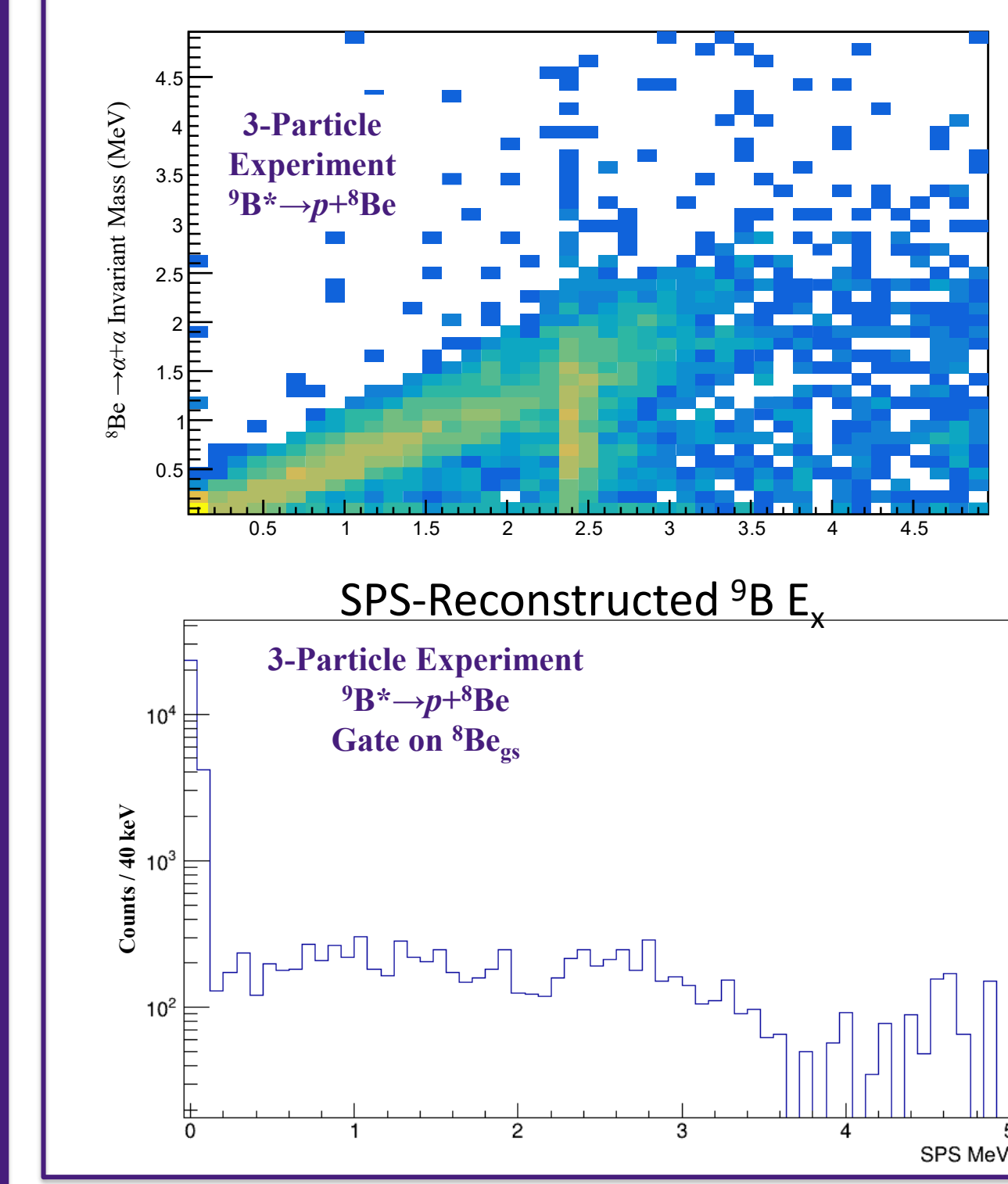
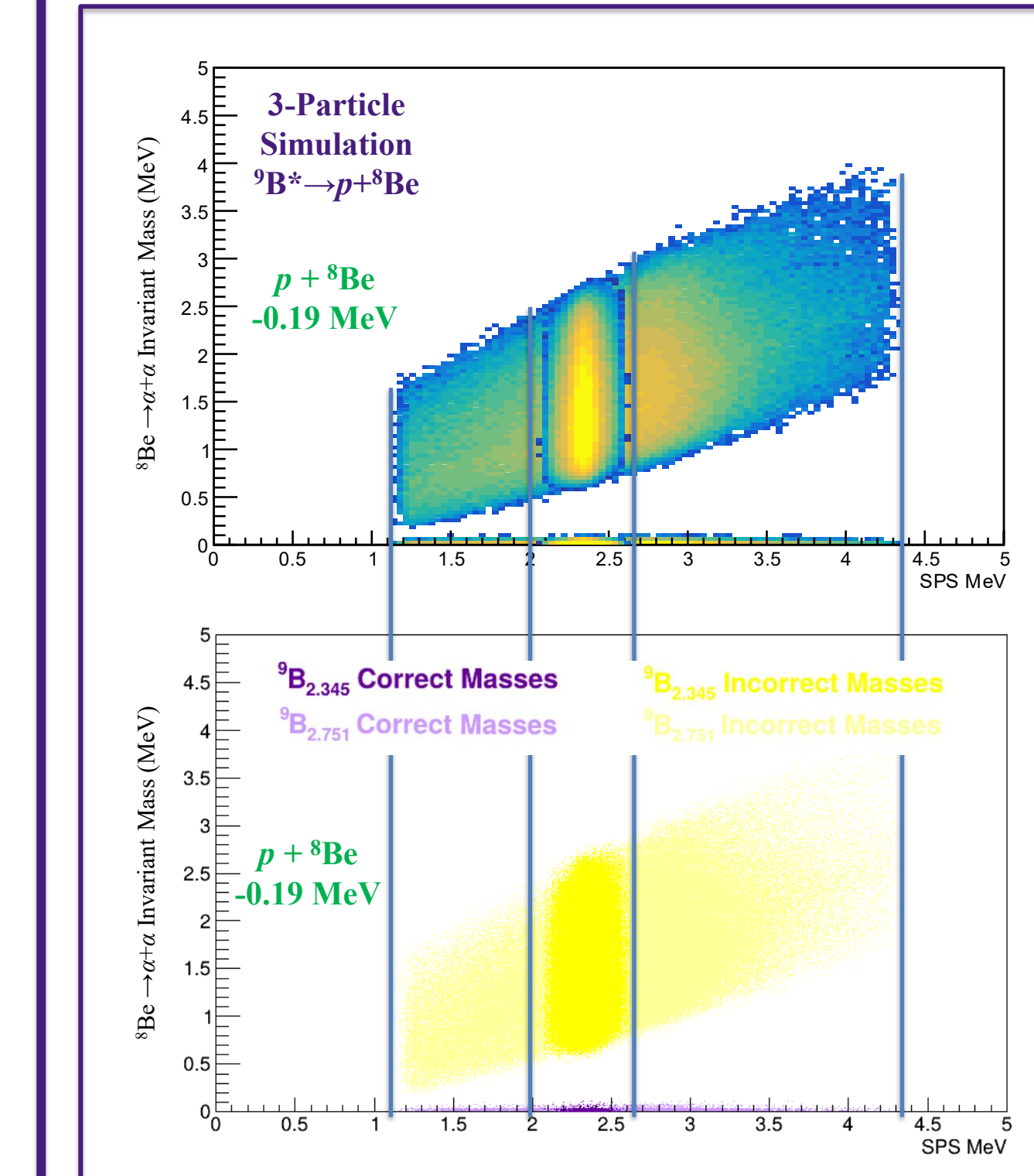


Figure 10 (above): Excitation energy of the assumed intermediate decay plotted against the SPS-reconstructed excitation energy of the ^9B recoil for experimental data. Invariant mass analysis was conducted for ^9B and decaying through $^9\text{B}^* \rightarrow p + ^8\text{Be}_{\text{gs}} \rightarrow \alpha + \alpha$ (left) and $^9\text{B}^* \rightarrow \alpha + ^5\text{Li}_{\text{gs}} \rightarrow p + \alpha$ (right). A 1D spectrum of SPS-reconstructed ^9B E_x is shown only for events where the assumed intermediate decay particle is within the expected range for ^8Be or ^5Li ground states.

Future Work

- Reconcile spectra with simulations and determine energies and widths for resonances in ^6Li and ^9B
- Conduct another experiment using the LSU Gas Target 2.0 with radioactive ^7Be beam to populate resonances in ^9B with $(^3\text{He}, p)$ reaction
- Investigate Dalitz plots and multi-body correlations

References

- [1] Tilley, D.R. et al., Nucl. Phys A 708, 3 (2002) [2] R. L. Dixon et al., Nucl. Phys. A 156: 1 33-42 (1970)
[3] Good, E.C. et al, Nucl Instrum. Methods Phys. Res. A 1003 (2021) [4] Tilley, D.R. et al., Nucl Phys A 745, 155-362 (2004)